

UAEU جامعة الإمارات العربية المتحدة
United Arab Emirates University

AI POWERED DIGITAL TWIN FOR RENEWABLE ENERGY

Presenter
Amulya Viswambharan
Researcher at UAEU

Outline

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- History
- Simulation Vs Digital Twin
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Introduction

- The global energy sector is undergoing major transformations to address critical challenges such as energy efficiency, sustainability, and supply security.
- Rising electricity demand and carbon reduction targets necessitate innovative strategies for optimizing current infrastructure and planning future expansions.
- Advanced digital technologies have emerged as key enablers in navigating these complex energy transitions.
- Among them, the Digital Twin (DT) concept stands out as a transformative solution, offering real-time insights, predictive capabilities, and intelligent control of energy systems.

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Digital Twin Definition

NASA

Digital twins—real-time virtual replicas of physical objects, systems, and processes—are surging to the forefront everywhere, from the doctor’s office and the manufacturing floor to the innards of cars, aircraft and rocket engines

Siemens

A digital twin is a virtual model mirroring a real-world object or system using sensor data and simulations. It allows real-time monitoring and analysis of a physical assets' behavior without having to touch the physical asset

IBM

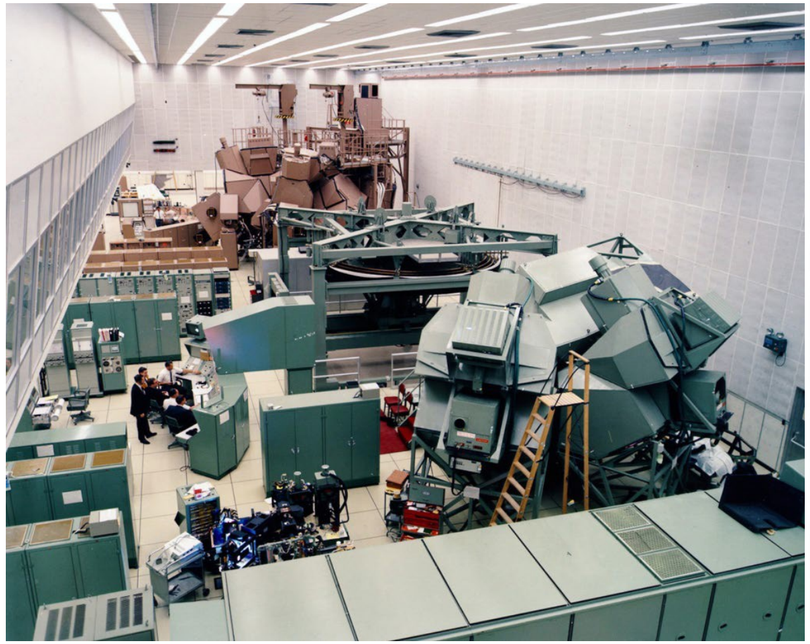
A Digital Twin is a virtual model designed to accurately reflect a physical object, system, or process. It helps organizations run simulations, analyze performance issues, and generate improvements

NVIDIA

Digital twins are virtual representations of products, processes, and facilities that enterprises use to design, simulate, and operate their physical counterparts

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The First Digital Twin: Apollo 13







- 15 simulators were used to train astronauts and mission controllers
- ➢ Simulator ➞ digital twin?
- Adapted to match conditions of actual spacecraft
- High fidelity model used to explore solutions and predict results

Image credit: NASA

Simulation Vs Digital Twin

Definitions

- Simulation: A model-based computation technique that replicates the behavior of a system under various conditions using predefined inputs.
  One-way, scenario-based and offline.
- Digital Twin: A dynamic, real-time virtual representation of a physical system that continuously updates using live sensor data and interacts with the real world.
  Two-way, continuous, connected, and intelligent.

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Feature	Simulation	Digital Twin
Connection to Physical System	Disconnected	Connected in real-time
Data Input	Static or batch data	Real-time sensor
Data Interaction	One-way	Two-way (actuation + feedback)Scope
Time-bound	scenario analysis	Continuous lifecycle management
Update Frequency	On demand, manually updated	Automatically updated via IoT
Purpose Design	planning, one-time validation	Monitoring, optimization, prediction, control
Example Use	Simulate load flow in a grid	Monitor and optimize a real PV plant in real time

Applications of Digital Twin



Energy



Manufacturing Process



Healthcare Services



Automotive Industry



Urban Planning



Structures and their systems

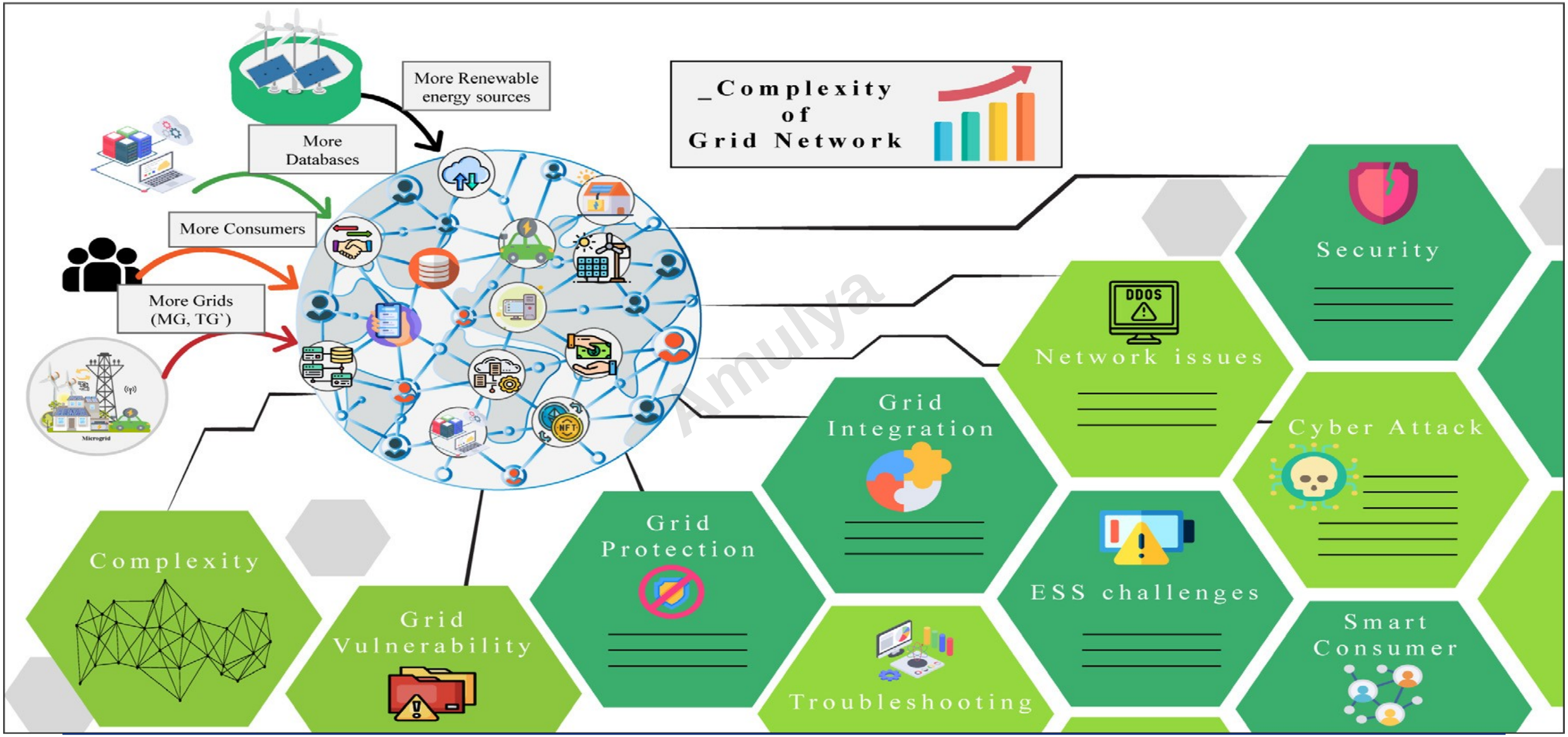
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Energy Digital Twin

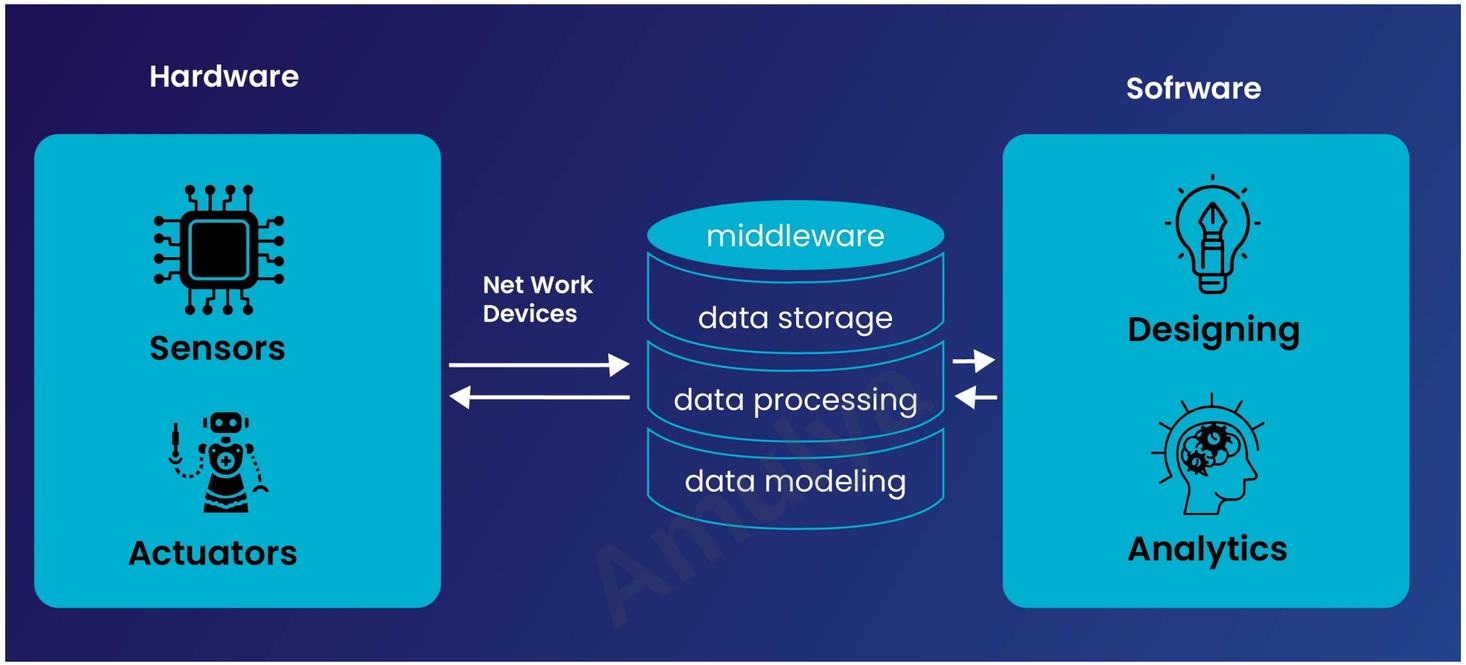
- Digital twins are virtual models of the real world that can be utilized in the planning designing, construction and operational phases of energy infrastructures
- Digital Twins use vast amount of data from various sources
- Digital twins can provide real-time insight and predictive prescriptive analysis
- Digital twins also enable the remote monitoring diagnostics control and automation of assets
- Digital twins can also provide scenario planning and insight when it comes to integrated resource planning of power systems

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Why Digital Twin in Energy

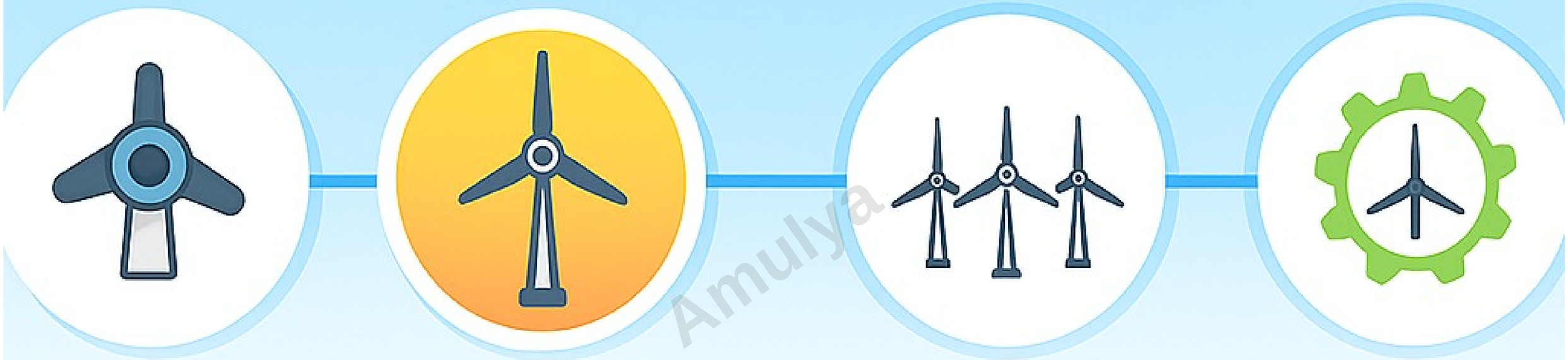


Core Architecture of Energy Digital Twin



- **Physical Layer:** Sensors, actuators, smart meters
- **Data Layer:** SCADA, IoT gateways, communication protocols (MQTT, OPC-UA)
- **Model Layer:** Physics-based + AI/ML-based models
- **Analytics Layer:** Edge/cloud computing, AI algorithms
- **Application Layer:** Dashboards, optimization tools, control feedback

Types of Digital Twin



Component or Part Twins

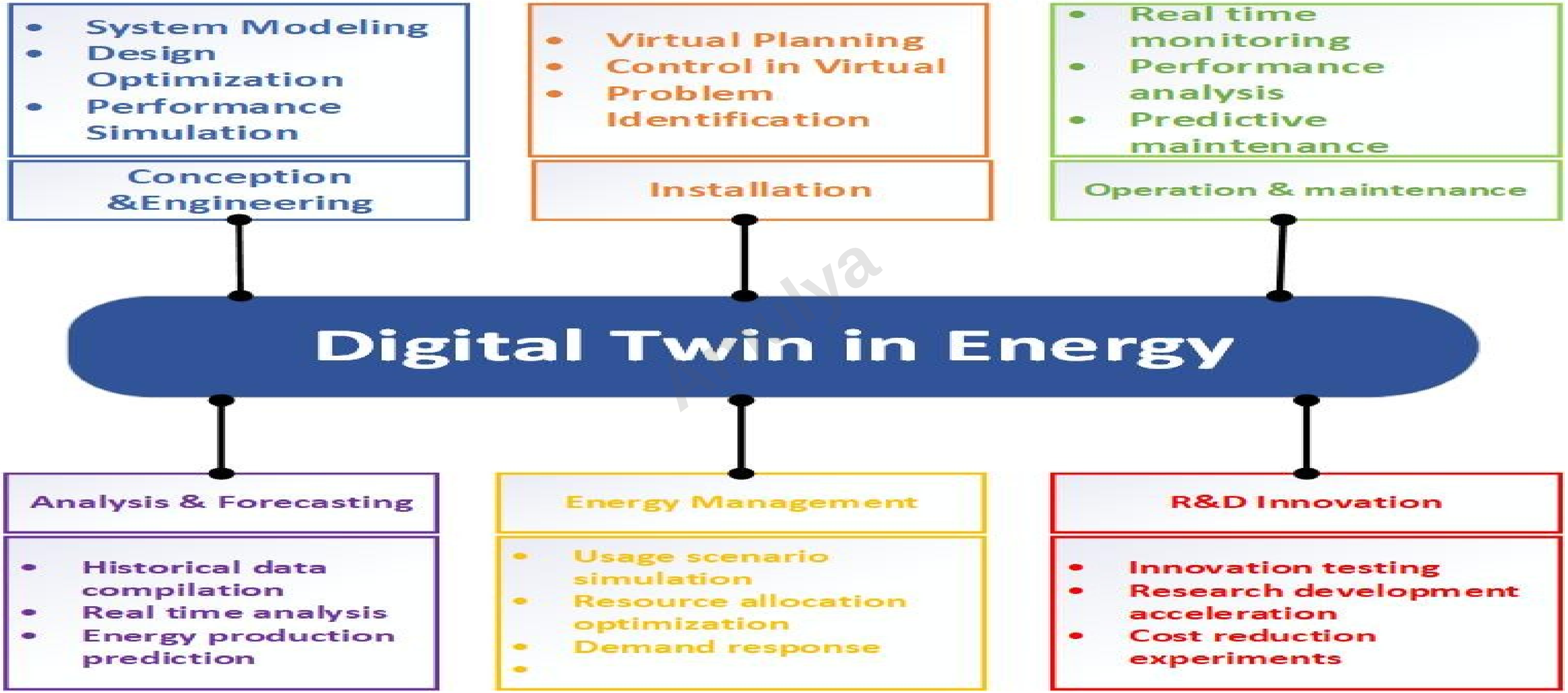
Asset Twins

System or Unit Twins

Process Twins

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Applications in Energy Systems



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Image credit: [1]

Digital Twin Process

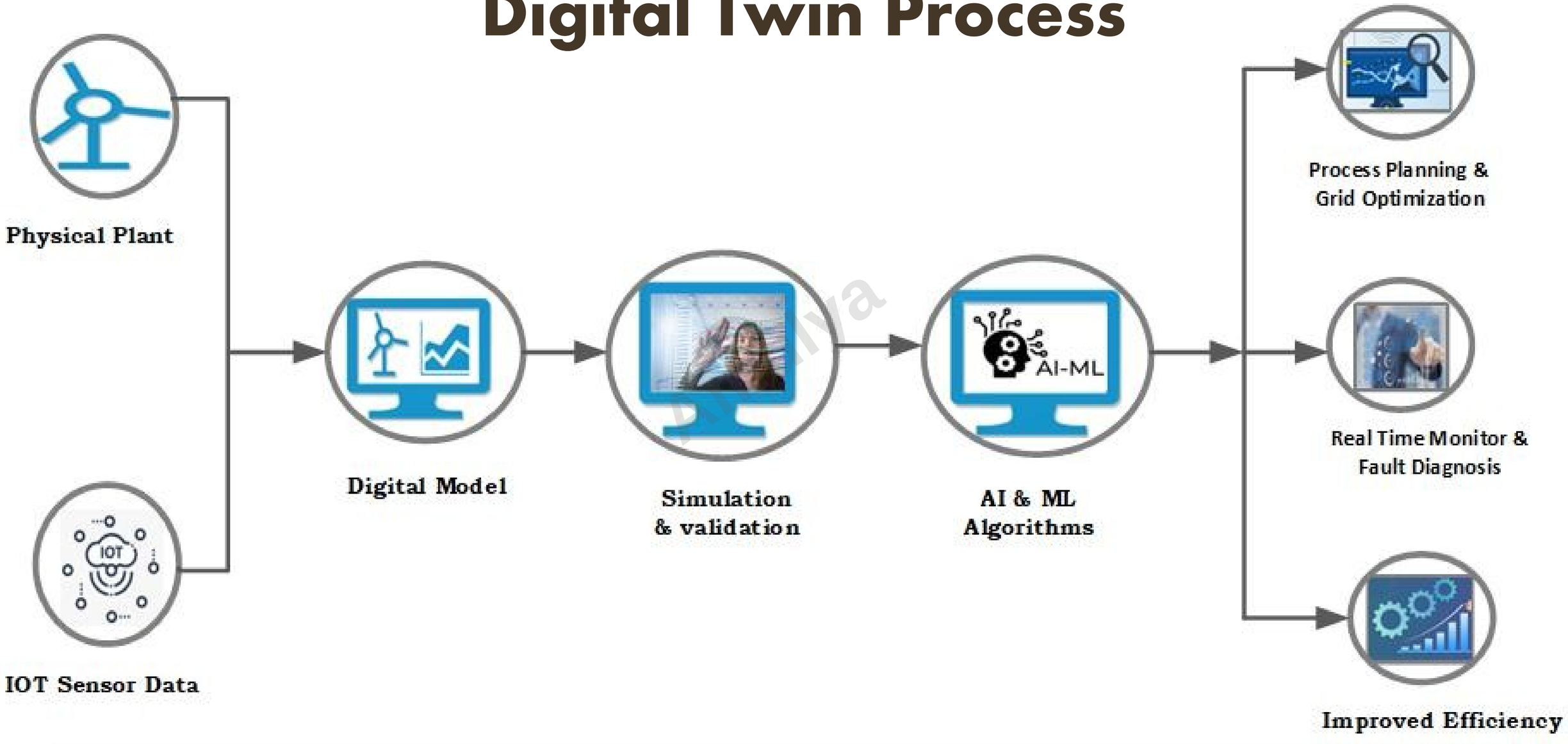


Image credit: [5]

Case Study - DT Implementation for PV Plant

➢ Objective

Enhance operational efficiency, predictive maintenance, and real-time performance insights

➢ Methodology

- IoT-enabled data acquisition
- Virtual twin creation
- AI/ML integration
- Scenario simulation & optimization

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Case Study - DT Implementation for PV Plant



1. Data Acquisition & Integration



1.1 IoT Sensor Network

Sensors monitor: temperature, humidity, irradiance, energy output, equipment health

Real-time, distributed architecture with secure database

Communication: MQTT, LoRaWAN, or ZigBee protocols



1.2 Data Integration

Combine IoT data with external sources: weather, environmental, grid data

Use of data fusion for harmonization

Enables real-world accuracy in virtual modeling

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Case Study - DT Implementation for PV Plant

2.Virtual Model & AI Integration

2.1 Virtual Model Creation

- Uses simulation tools to mimic layout, electrical/thermal behavior, tracking systems
- Real-time calibration with plant data
- Tests impacts of irradiance, temperature, wind, etc.

2.2 AI/ML Integration

- ML models trained on historical & live data
- Tasks: anomaly detection, fault prediction, efficiency forecasting
- Continuous learning improves predictive accuracy

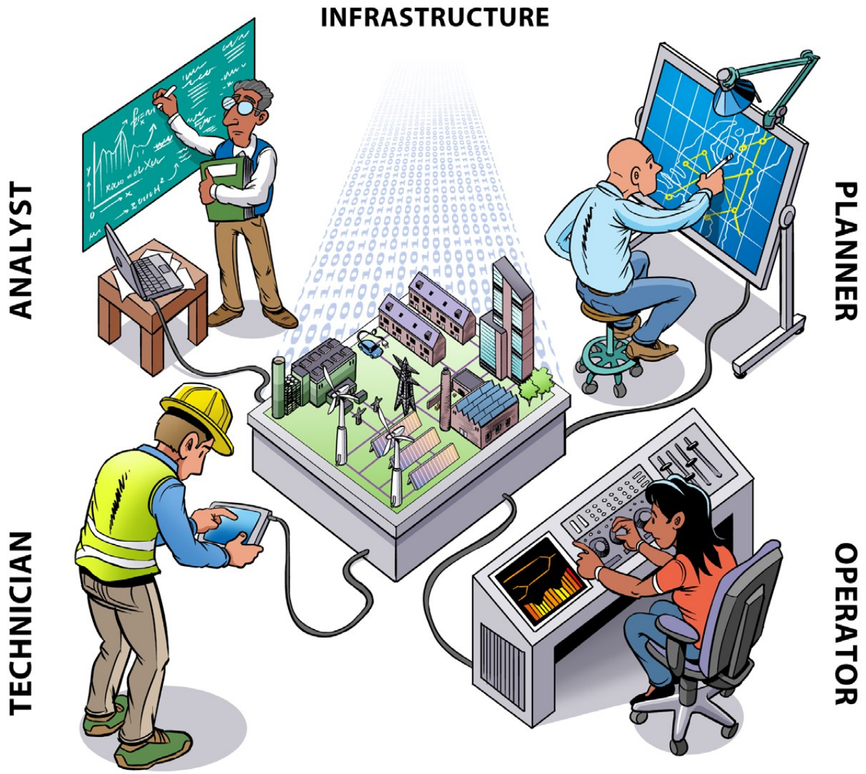


Fig.1 Role-Based Engagement in Digital Twin-Enabled Infrastructure Systems[9]

Case Study - DT Implementation for PV Plant

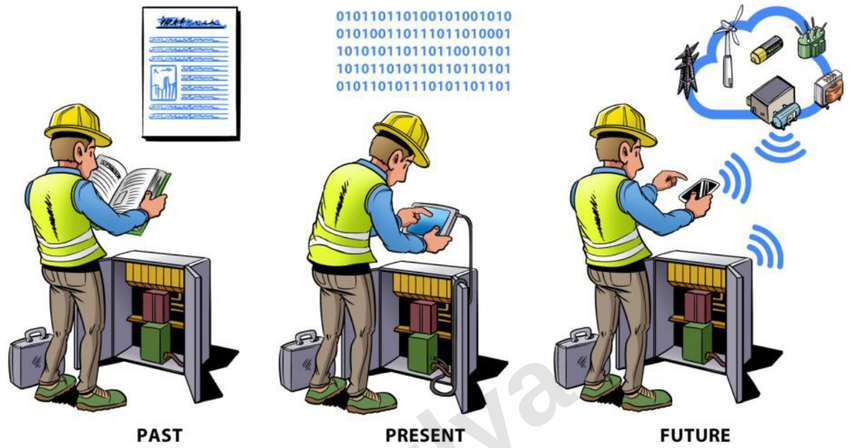


Fig: 2 From Manual Logs to Digital Twin Intelligence[9]

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3. Real-Time Monitoring & Predictive Maintenance

3.1 Monitoring Interface

- Dashboards for KPIs: energy output, temp, equipment health
- Customizable views: graphs, heatmaps, alerts
- Historical data analysis for strategy refinement

3.2 Predictive Maintenance

- Early anomaly detection using ML
- Triggers maintenance before failures
- Benefits: less downtime, longer equipment life, cost savings

Case Study - DT Implementation for PV Plant

4. Scenario Simulation & Optimization

4.1 Virtual Scenarios

- Simulate weather events, malfunctions, degraded modules
- Analyze impact on performance and reliability
- Stress testing for extreme UAE climate

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4.2 Optimization Algorithms

- Run what-if simulations to find best operating configurations
- Dynamic adjustment for real-time optimization
- Outcomes: higher yield, lower cost, better sustainability

Standards & Tools

- **ISO/IEC 30173** – Digital Twin Standard
- **IEEE P2806** – Asset Representation
- **Tools:** MATLAB Simulink, Azure DT, Mindsphere, OpenDSS

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Benefits & Challenges



BENEFITS:



ENHANCED ASSET PERFORMANCE



REDUCED DOWNTIME



DATA-DRIVEN DECISION MAKING



CHALLENGES:



HIGH DEPLOYMENT COST



INTEGRATION WITH LEGACY SYSTEMS



MODEL ACCURACY AND CALIBRATION

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Future Research Directions

- Smart Grid Optimization & Predictive Maintenance
- Renewable Energy Integration & Forecasting
- Demand Response & Customer Behavior Modelling
- Electric Mobility & Battery Innovation
- Cybersecurity, Communication Protocols & Regulatory Standards

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Conclusion

- Digital Twin provides a powerful bridge between physical energy systems and digital platforms, enabling real-time performance monitoring and control.
- It enhances system efficiency, fault detection, predictive maintenance, and lifecycle management across various energy applications.
- Supports the renewable energy transition by improving sustainability, safety, and environmental impact through advanced modeling and simulation.
- Enables smarter energy management through better data visualization, interactivity, and system-wide optimization.
- **Decarbonization** and **Digitalization** emerge as the twin pillars driving the future of energy systems — both enabled and accelerated by Digital Twin technology.

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THANK YOU

Presented By Amulya Viswambharan